

GRIFOLS

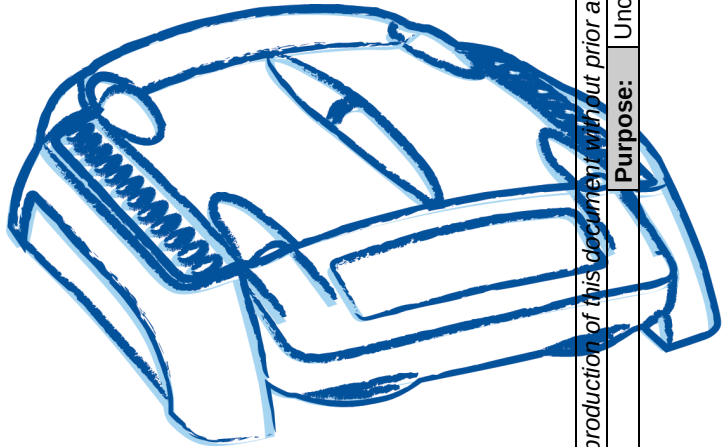
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DGtherm

3044105
03/2016

GRIFOLS

DGtherm
Instructions for Use



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DG THERM

Instructions for Use



GRIFOLS

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PLEASE READ THESE INSTRUCTIONS FOR USE CAREFULLY
BEFORE WORKING WITH THE DG THERM INCUBATOR

Information about incubator safety can be found in Section 2.

The symbols used in these Instructions for Use are as follows:

	WARNING: Indicates the possibility of personal injury, loss of life, or harm to the environment if the instructions are not followed.
	CAUTION: Indicates the possibility of: 1) instrument damage if instructions are not followed; or 2) compromising of samples or sample integrity due to cross-contamination.
	NOTE: Used for clarifications or additional or emphasized information.

The defined terms used in these Instructions for Use are as follows:

Operator:	The person operating the equipment for its intended use.
Supervisor (or Responsible Body):	The individual or group responsible for the use and maintenance of the equipment and for ensuring that operators are adequately trained.
Qualified Technician:	Person responsible for the installation, reparation and special maintenance of the equipment who has received specific training for this purpose.

All restrictions referred to Supervisor are valid for Operator.

Equipment design and specifications are subjected to change without previous notification.

This document is available in various languages. Translations have been carried out from the master document in English. In case of doubt or discrepancies, what is expressed in the master document in English will be given priority.

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1 Introduction

THIS INCUBATOR IS INTENDED FOR
IN VITRO DIAGNOSTIC USE

These Instructions for Use are intended for the users of the DG THERM Incubator and contain all the information necessary to safely and adequately work with the incubator.

Carefully read these Instructions for Use before you work with the incubator.

If you have any questions, please contact Technical Service before starting any operation.

These Instructions for Use should be accessible to all personnel who work with the incubator.

1.1 Intended Use

The DG THERM Incubator is a device specifically designed to incubate DG Gel cards and/or sample tubes at 37 °C for immunohematology applications.

1.2 Principle of Operation

The DG THERM Incubator incubates DG Gel cards and/or sample tubes in a dry heating bath at 37 °C for 15 minutes. The DG THERM Incubator has two independent timers and can carry out two independent incubation batches.

1.3 Product Limitations

The DG THERM Incubator is designed only to incubate DG Gel cards and/or sample tubes as described in Section 1.1 and should not be used for any other type of test.

1.4 Operator Training

The use of the DG THERM Incubator does not require specific training beyond these Instructions for Use and the training established by the laboratory.



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2 Safety Information

Grifols takes no responsibility for unauthorized modifications of the equipment or for failure of operating personnel to heed the warnings and cautions outlined in these Instructions for Use.

2.1 Warnings and Cautions

Before operating the equipment, review and become familiar with these warnings and cautions. Adherence to the warnings and cautions will help protect you, your colleagues, the equipment, the test results, and ultimately your patients.

2.1.1 Health and Safety Notices



WARNING: Use the incubator only for the operations described in these Instructions for Use.



WARNING: Unless specifically instructed to do so, do not drop or put anything into any opening in the incubator.



WARNING: Do not use the incubator if it is not working properly or if it is damaged. Examples of damage include:

- Damage caused by dropping the incubator.
- Damage caused by spilling liquid on or into the incubator.
- Damage caused by transport or storage under conditions outside of the transport or storage conditions listed in Section 3.1 or severe stresses during transport.
- Damage to the power cord or its plug.



WARNING: Do not use the incubator in the presence of flammable gases or volatile products. Do not put flammable, explosive, or chemical materials into the incubator.



WARNING: Inappropriate use of this incubator can damage the safety features of this incubator.



WARNING: Chemical and Biological Hazards: Appropriate safety precautions, including Universal Precautions, must be strictly followed, as well as local, state, and national regulations. Appropriate personal protective equipment must be worn at all times when working with liquids in and around the equipment. Dispose of all chemical and biological waste, including empty sample and reagent containers or tubing, in accordance with local, state, and national regulations and practices. In the event of a spill of sample or reagent material, follow all site protocols for spill response, control, and disposal.



WARNING: Follow Universal Precautions when working with potentially hazardous materials, including the following circumstances:

- When handling samples. Treat these as potentially infectious material.
- When working with liquids in and around the equipment.
- When handling waste and waste containers. They may contain biohazardous waste.
- When performing maintenance activities.

Always wear appropriate protective equipment when working with hazardous materials.



WARNING: Any part of the incubator that comes into contact with blood, serum samples, or any other biological liquid should be treated as potentially infectious.



CAUTION: Do not use accessories which are not supplied or recommended by the manufacturer.



CAUTION: The incubator may only be disassembled by qualified personnel. Before disassembling, the incubator must be unplugged.

2.1.2 Electrical Safety Notices



WARNING: To avoid risk of electric shock, the incubator should be connected to a grounded power outlet.



WARNING: Electromagnetic compatibility and electrical safety tests have been performed using the power cord supplied by the manufacturer. Do not use a power cord other than the power cord provided with the incubator as this may result in increased electromagnetic emissions and/or compromise the incubator.



WARNING: If you spill cleaning solution or other liquids into the incubator, the incubator must be unplugged, cleaned, and decontaminated. The incubator must only be disassembled by qualified personnel and it must not be plugged in again until it has been fully inspected by qualified personnel.

2.1.3 Installation Notices



WARNING: Do not let the incubator or its power cord come into contact with hot surfaces.



CAUTION: Do not place the incubator outdoors or where it could get wet.

2.1.4 Operational Notices



CAUTION: To handle DG Gel cards, follow the DG Gel Instructions for Use.



CAUTION: Times shown by the timer of the DG THERM Incubator (incubation time and time exceeded) are correct if the operator has worked as described in Section 6; in other words without delays due to deviation from the procedure.



CAUTION: When stopping a flashing timer at the end of an incubation batch, make sure to press the appropriate **Start/Stop Timer** button (the **P1** button or the **P2** button) to stop the timer that is flashing.



CAUTION: For the card elevator system to work properly, the trays must be completely and correctly loaded into the incubator.



CAUTION: To avoid overincubation or underincubation of the cards and/or sample tubes, follow the procedures described in Section 6.



CAUTION: Cards and/or sample tubes should be removed immediately after the end of incubation. After the timer stops, the thermal block continues to heat and so the cards and/or sample tubes continue to be incubated if they are not removed.



CAUTION: When incubation ends, the timer displays the pre-programmed time of 15 minutes.



CAUTION: If incubation stops before completion, discard the DG Gel cards and invalidate the test results.



CAUTION: Pressing and holding the appropriate **Start/Stop Timer** button (the **P1** button or the **P2** button) during an incubation batch stops the incubation timer, but the cards and/or sample tubes continue incubating until they are removed from the incubator.



CAUTION: After an incubation batch with a modified incubation time has finished, the timer setting returns to the pre-programmed time of 15 minutes.



CAUTION: If the DG THERM Incubator displays an error code, turn the incubator off and then back on again. If the error symbol appears again, contact Technical Service and give them the error code.



CAUTION: To stop the alarm, press the **P1** button or the **P2** button. If the alarm continues, contact Technical Service.



CAUTION: Repairs should only be carried out by qualified personnel.

2.1.5 Maintenance Notices

- **WARNING:** Before disassembling, cleaning, or decontaminating the incubator (as described in Section 7), unplug the incubator from the power supply.
- **WARNING:** During cleaning and/or decontamination, wear protective gloves.
- **WARNING:** Some cleaning or decontamination products, such as sodium hypochlorite solution, may be corrosive, irritating for the skin and eyes, or toxic if inhaled, absorbed, or ingested.
- **CAUTION:** Before cleaning and/or decontaminating the incubator, remove all cards and/or sample tubes from inside the incubator.

2.1.6 Transport and Storage Notices

- **WARNING:** Decontaminate the incubator before transport and/or storage.
- **CAUTION:** Use only the original packaging to store or transport the incubator.

2.1.7 Disposal Notices

- **WARNING:** Only qualified personnel should disassemble this incubator.
- **WARNING:** The incubator contains electronics. Electronics can contain hazardous materials. Dispose of all electronics in accordance with local, state, and national regulations.

2.2 Incubator Labeling

The DG THERM Incubator comes with the following labels:

- Packaging Label.
- Product Label.
- Biological Risk Level Label.

The following symbols are used on the labels:



Serial number



Catalog number



Temperature range permitted during storage and transport



Permitted relative humidity range during storage and transport



In vitro diagnostic medical device



Manufacturer



Consult Instructions for Use



Biological risk, infectious substance



Date of manufacture



DataMatrix including the following information:

(01) GTIN number

(11) Manufacturing year

(21) Serial number of the instrument



Safety mark that certifies that the instrument meets the applicable USA and Canadian safety standards.

Packaging Label

The Packaging Label (Figure 1) is located on the right upper part of the front of the box for the DG THERM Incubator.

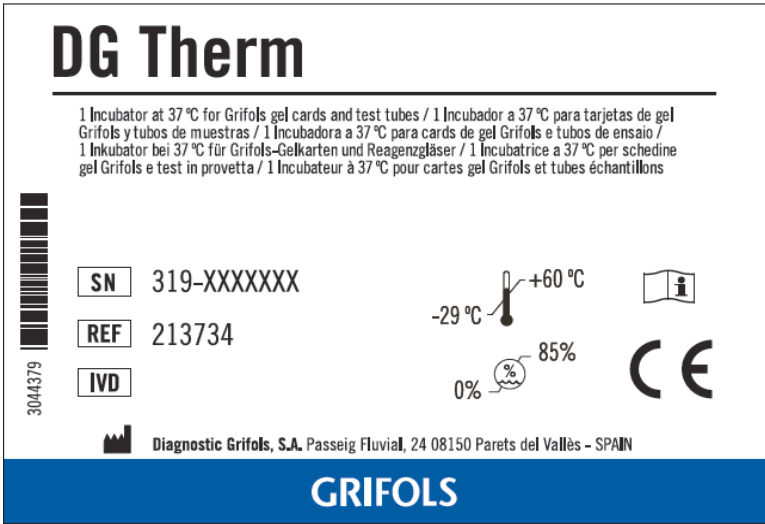


Figure 1. Packaging Label

Product Label

The Product Label (Figure 2) is located on the rear of the incubator and contains technical information (voltage, frequency, and input power), as well as the model name, catalogue number, serial number, manufacturing date, name and address of the manufacturer, and regulatory information.



Figure 2. Product Label

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Biological Risk Level Label

The Biological Risk Level Label (Figure 3) is located on the flap of the trays (Figure 4, no. 5) and the thermal block (Figure 6, no. 1), and warns the operator about potential biological risks caused by residual substances deposited on the surface of the incubator and especially on the trays.



Figure 3. Biological Risk Level Label



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3 Technical Specifications and Accessories

The following sections contain information on the technical specifications and accessories available for the DG THERM Incubator.

3.1 Technical Specifications

Table 1 lists the technical specifications for the incubator.

Table 1. Technical Specifications

Feature	Technical Specifications	
MODEL	DG THERM Incubator	
LOAD CAPACITY	24 DG Gel cards. 24 sample tubes.	
INCUBATION BATCHES	2 batches with independent timer	
INCUBATION TEMPERATURE	37 °C ± 1 °C	
TEMPERATURE UNITS	Programmable, °C or °F	
INCUBATION TIME	15 minutes, modifiable in steps of 5 minutes to 60 minutes	
PREHEATING TIME	Approximately 6 minutes	
TEMPERATURE SENSORS	Yes, two independent sensors to measure and control temperature	
ALARM	Yes, to indicate end of incubation and error situations	
VOLUME OF ALARM	Programmable, 3 levels	
INTERFACE	LCD screen and buttons	
POWER REQUIREMENTS	Voltage:	100-240 V ~
	Frequency:	50-60 Hz
	Input power:	270 W

Feature	Technical Specifications	
PROTECTION AGAINST ELECTRIC SHOCK	Class:	I
INSTALLATION CATEGORY	Overvoltage category II (local level, appliances, portable equipment, etc.)	
DIMENSIONS	33 cm deep by 31 cm wide by 10 cm high	
WEIGHT	Approximately 4 kg	
OPERATING CONDITIONS	Indoor use	
	Temperature:	15 °C to 30 °C
	Maximum relative humidity non-condensing:	80%
	Maximum altitude:	3000 m
	Power supply maximum voltage fluctuations:	±10% of nominal voltage
TRANSPORT AND STORAGE CONDITIONS	Temperature:	-29 °C to 60 °C
	Maximum relative humidity non-condensing:	85%
PROTECTION AGAINST OVERHEATING	50 °C ± 5 °C	
USEFUL LIFE	5 years (based on an average use of 8 hours/day and 200 days/year)	

3.2 Accessories

Table 2 lists the accessories available for the incubator.

Table 2. Accessories

PART #	DESCRIPTION	QUANTITY
213732	DG THERM Trays	6
230881	PT100 temperature probe for DG Gel cards	1
230879	Digital thermometer for PT100 temperature probes	1



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4 Description

Figures 4 and 5 respectively show a front and rear view of the DG THERM Incubator.

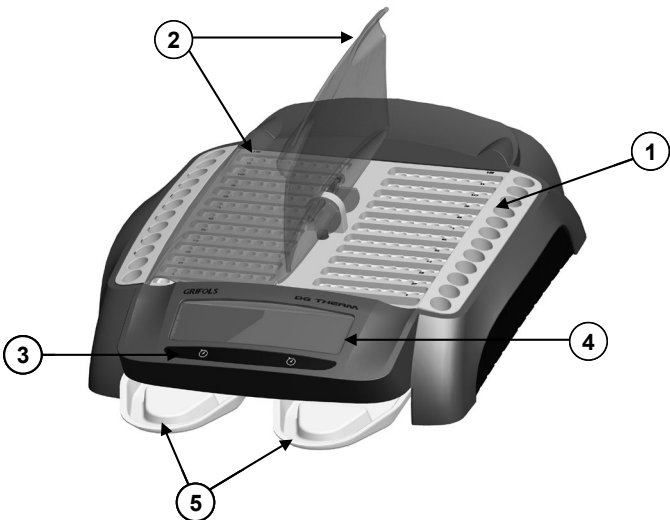


Figure 4. Front View of DG THERM Incubator

- (1) Thermal block
- (2) Covers
- (3) **Start/Stop Timer** buttons **P1** and **P2**
- (4) Display
- (5) Trays

Description

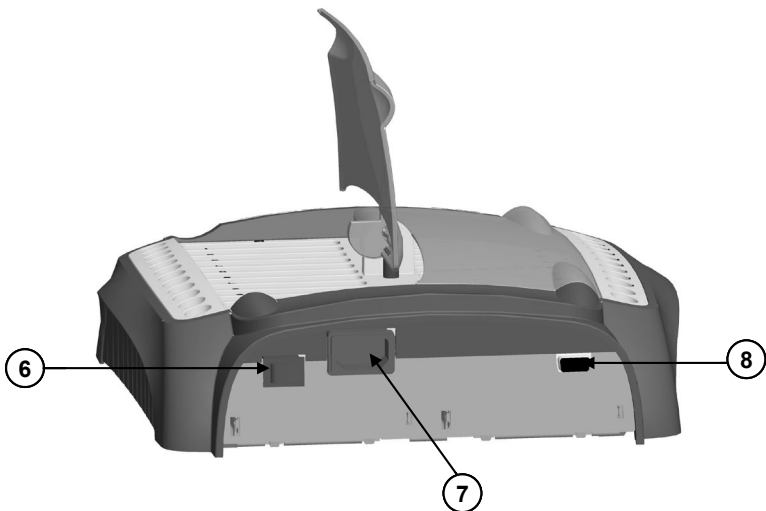


Figure 5. Rear View of the DG THERM Incubator

- (6) Power switch
- (7) Power cord connection
- (8) Computer connection (for the exclusive use of Technical Service)

4.1 Thermal Block

The thermal block (Figure 6) holds cards and/or sample tubes for incubating.

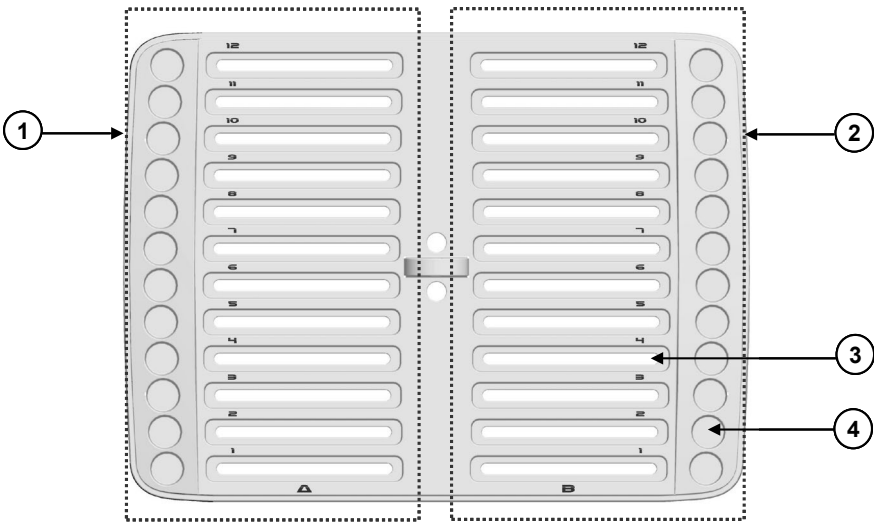


Figure 6. Thermal Block

- (1) Incubation area A
- (2) Incubation area B
- (3) Slots for DG Gel cards
- (4) Slots for sample tubes

The thermal block is divided into two incubation areas (Figure 6, no. 1 and no. 2): incubation area A and incubation area B. Each area holds a maximum of 12 DG Gel cards (Figure 6, no. 3) and 12 sample tubes (Figure 6, no. 4) that are aligned with the cards.

4.2 Incubator Covers

The DG THERM Incubator has two covers (Figure 4, no. 2) that keep the cards down for optimum incubation.

4.3 Start/Stop Timer Buttons

The DG THERM Incubator has two **Start/Stop Timer** buttons (Figure 4, no. 3): the **P1** button (Figure 7, no. 1) and the **P2** button (Figure 7, no. 2). You can start and stop each timer independently.



Figure 7. Start/Stop Timer Buttons

- (1) **Start/Stop Timer** button P1
- (2) **Start/Stop Timer** button P2

4.4 Display

The display (Figure 8) shows time remaining to the end of incubation and the temperature of the thermal block. The display also shows possible error symbols.

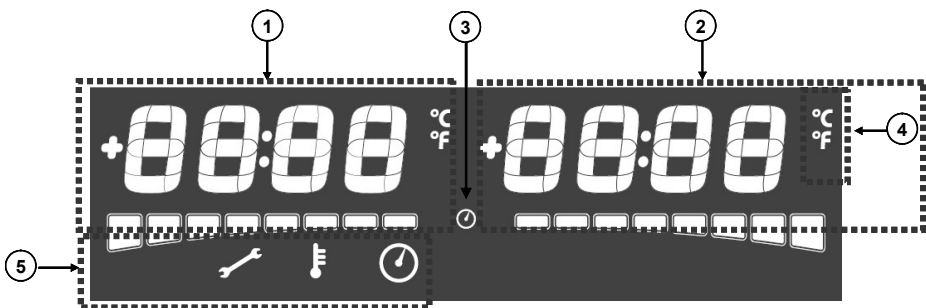


Figure 8. Display

- (1) Timer 1 information
- (2) Timer 2 information
- (3) Timer measurement indicator
- (4) Temperature units
- (5) Error symbols area

4.4.1 Timer 1 and 2 Information

The DG THERM Incubator has two independent timers (Figure 9, no. 1 and no. 2).

 **NOTE:** You can use either timer for either incubator area. However, it is recommended that you use timer 1 for incubations carried out in incubation area A and timer 2 for incubations carried out in incubation area B.

In normal mode, the display shows a digital counter and a progress bar for each timer. The progress bar fills in as the pre-programmed time elapses. If you do not stop the timer after incubation has finished, the timer flashes the amount of time past the end of incubation, preceded

by the **Time Exceeded** symbol  (Figure 10, no. 5).



Figure 9. Display During Incubation

- (1) Timer 1 information
- (2) Timer 2 information
- (3) Timer 1 progress bar
- (4) Timer 2 progress bar

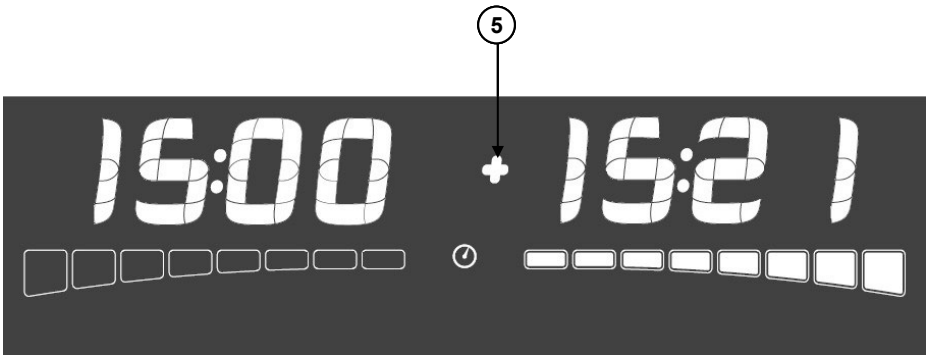


Figure 10. Display Showing Time Beyond End of Incubation in Timer 2

(5) Time Exceeded symbol

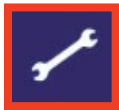
4.4.2 Display Temperature Units

The DG THERM Incubator can display incubation temperature in °C or in °F. To change the temperature units displayed, see Section 5.4.

4.4.3 Error Symbols Area

The error symbols area is located on the left lower part of the display (Figure 8, no. 5) and displays symbols for various error situations that occur during start-up or while incubating a batch. For more information, see Section 12.1.

The following symbols may be displayed in the error symbols area:



Repair symbol: Contact Technical Service



Temperature Out-of-Range symbol: Incubation temperature is out of the specified range



Time Out-of-Range symbol: Batch completed with incorrect incubation time (Timer 1 and/or 2)

4.5 Trays

The trays (Figure 4, no. 5) are located in the lower part of the incubator. They are used to contain any spills during incubator use. The trays are removable and disposable. The trays also form part of the card elevation system of the thermal block.

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5 Installation

The following sections contain instructions on installing the DG THERM Incubator, as well as information about changing the settings.

5.1 Installation Requirements



WARNING: To avoid risk of electric shock, the incubator should be connected to a grounded power outlet.



WARNING: Do not let the incubator or its power cord come into contact with hot surfaces.



CAUTION: Do not place the incubator outdoors or where it could get wet.



NOTE: If the incubator has been stored in environmental conditions different from operating conditions, keep the incubator in operating conditions for at least one hour before plugging it in. For more information on operating conditions, see Section 3.1.

The installation requirements for the incubator are as follows:

- A minimum space of 40 cm wide by 56 cm deep (includes space to connect the power cord).
- Do not position the incubator so that it is difficult to operate the power switch (Figure 5, no. 6) or access the power cord.
- Do not place anything on top of the incubator.
- Connect the incubator to a power supply which meets the power supply requirements listed in Section 3.1 (including ground protection).

5.2 Unpacking the Incubator

After you choose an appropriate location to install the incubator, unpack the incubator by following these instructions:

1. Open the upper part of the box.
2. Remove all packing protection and accessories.
3. Remove the protection bag and place the incubator in the chosen position.
4. Make sure that all elements specified in the attached *Packing List* are included.
5. Check that the incubator has not been damaged during storage or transport.

5.3 Installing the Incubator

To install the DG THERM Incubator, follow these instructions:

1. Ensure that the DG THERM Trays are completely inserted into the incubator and correctly placed.
2. Ensure that the covers open correctly.
3. Ensure that the card elevator system works as described in Section 8.4.
4. Connect the power cord to the incubator (Figure 5, no. 7) and to a grounded power outlet.
5. Turn on the incubator (Figure 5, no. 6).
6. Make sure that the display switches on and shows the preheating phase symbols “ --:-- ”.
7. After the preheating phase has completed, make sure that the display shows the pre-programmed time of 15 minutes for both timer 1 and timer 2.

The DG THERM Incubator is supplied with a conformity certificate stating that the temperature of the thermal block (37 °C) has been verified using instruments calibrated by accredited entities. The DG THERM Incubator does not need to be recalibrated during its useful life if it is used under normal working conditions.

5.4 Changing Settings

The settings for the incubator have been preset:

- Alarm volume: High.
- Temperature units: °C.
- Date and time: Greenwich Mean Time (GMT) time zone.



NOTE: The Date and Time setting is used to maintain traceability in a “log file” with relevant information and possible incidences that have occurred during incubation. Only Technical Service personnel can access and use this file.

To change these settings, follow these instructions:

1. Turn the power switch off.
2. Turn on the power switch while you press and hold the **P1** button.

The DG THERM Incubator starts in **Configuration Mode** and the display shows the alarm volume menu. Three settings are available:

- VOL 0: Deactivated.
- VOL 1: Medium volume.
- VOL 2: High volume.

3. Press the **P2** button to change the volume setting to the desired level.
4. Press the **P1** button to move to the temperature units menu. Two settings are available:
 - SEL °C: Temperature in degrees Centigrade.
 - SEL °F: Temperature in degrees Fahrenheit.
5. Press the **P2** button to change the setting to the desired temperature units.
6. Press the **P1** button to move to the date and time menu. The display flashes the date and time.
7. Press the **P2** button to modify the value of each field and the **P1** button to move onto the next field.



NOTE: Before changing settings, check with your supervisor.



NOTE: The date and time are preset to GMT.

8. Save the settings by pressing and holding the **P1** button until the alarm emits a long beep. The DG THERM Incubator shows the pre-programmed incubation time of 15 minutes.



NOTE: If no button is pressed for 15 seconds at any point while settings are being changed, the incubator leaves **Configuration Mode** and the changes to the settings are not saved.



NOTE: To save changes to the settings, press and hold the **P1** button until the alarm emits a long beep. Otherwise, the settings will not be saved.

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6 Operation



WARNING: Do not use the incubator if it is not working properly or if it is damaged. Examples of damage include:

- Damage caused by dropping the incubator.
- Damage caused by spilling liquid on or into the incubator.
- Damage caused by transport or storage under conditions outside of the transport or storage conditions listed in Section 3.1 or severe stresses during transport.
- Damage to the power cord or its plug.



CAUTION: To handle DG Gel cards follow the DG Gel Instructions for Use.

6.1 Incubating a Batch

To correctly use the DG THERM Incubator, follow these instructions:

1. Turn the power on (Figure 5, no. 6). Make sure that the display shows the symbols “--:--” and the **Temperature Out-of-Range** symbol  is flashing to indicate that the incubator is in preheating mode. After incubation temperature is reached, the **Temperature Out-of-Range** symbol  disappears, the alarm beeps, and timers 1 and 2 will show the pre-programmed incubation time of 15 minutes.



NOTE: During the preheating phase, the timers cannot be started.

Do not load cards and/or sample tubes into the thermal block until the incubator has reached the incubation temperature and the timers show the pre-programmed incubation time of 15 minutes.

2. Open the cover of the incubation area (A and/or B) that you wish to use.
3. Load the cards into the card slots of the thermal block (Figure 6, no. 3) and/or load the sample tubes into the tube slots of the thermal block (Figure 6, no. 4). While the cover is open, the cards are lifted slightly to make loading easier.
4. Close the cover. The cards lower fully into the thermal block.
5. Press the **P1** button or the **P2** button to start the timer for the incubation batch.



NOTE: You can use either timer for either incubator area. However, it is recommended that you use timer 1 for incubations carried out in incubation area A and timer 2 for incubations carried out in incubation area B.

During incubation, the display shows the remaining incubation time and will gradually fill in the progress bar.

After incubation has completed, the alarm beeps and the display for the corresponding timer flashes.



CAUTION: Times shown by the timer of the DG THERM Incubator (incubation time and time exceeded) are correct if the operator has worked as described in Section 6; in other words without delays due to deviation from the procedure.

- Press the **P1** button or the **P2** button again to stop the timer that is flashing.



CAUTION: When stopping a flashing timer at the end of an incubation batch, make sure to press the appropriate **Start/Stop Timer** button (the **P1** button or the **P2** button) to stop the timer that is flashing.

If the timer is not stopped after incubation has finished, the display shows the amount of time that has exceeded the pre-programmed incubation time and flashes the **Time Exceeded**

symbol . The alarm also beeps.

- Open the cover for the incubation area that has finished incubating. The card elevation system raises the cards so that they can be removed more easily.
- Immediately remove the cards from the incubator.



CAUTION: Cards and/or sample tubes should be removed immediately after the end of incubation. After the timer stops, the thermal block continues to heat and so the cards and/or sample tubes continue to be incubated if they are not removed.

6.2 Displaying Incubation Temperature

To momentarily display the incubation temperature, press the **P1** button or the **P2** button during an incubation batch.

The display will show the temperature of the thermal block for a few seconds. After a few seconds, the temperature disappears and the time remaining reappears.



NOTE: The temperature displayed when you press either of the two **Start/Stop Timer** buttons (the **P1** button or the **P2** button) is the temperature of the entire thermal block; i.e., the temperature is the same for incubation area A and incubation area B.

6.3 Resetting the Timer

To reset the timer to the pre-programmed incubation time after an incubation batch has completed and the cards and/or sample tubes have been removed, press the **Start/Stop Timer** button (the **P1** button or the **P2** button) corresponding to the timer you want to reset.

6.4 Stopping Incubation

To stop incubation timing before the incubation time is finished, press and hold the appropriate **Start/Stop Timer** button (the **P1** button or the **P2** button) until the alarm emits several short beeps and a long beep. The display alternates between showing the time remaining when the

timer stopped and the **Time Out-of-Range** symbol .

 **CAUTION:** If incubation stops before completion, discard the DG Gel cards and invalidate the test results.

 **CAUTION:** Pressing and holding the appropriate **Start/Stop Timer** button (the **P1** button or the **P2** button) during an incubation batch stops the incubation timer, but the cards and/or sample tubes continue incubating until they are removed from the incubator.

6.5 Changing the Pre-Programmed Timer Setting

The DG THERM Incubator has been pre-programmed to incubate DG Gel cards and sample tubes that require incubation at 37 °C for 15 minutes (according to the corresponding Instructions for Use for DG Gel cards).

To change the pre-programmed timer setting for one of the two timers, follow these instructions:

1. Press and hold the **Start/Stop Timer** button (the **P1** button or the **P2** button, depending on which timer setting you want to change) until the alarm beeps and the corresponding timer flashes.
2. Press and hold the same **Start/Stop Timer** button again to increase the timer setting in 5 minute increments until you reach the desired timer setting (the maximum timer setting permitted is 60 minutes).
3. When the **Start/Stop Timer** button is released, the alarm emits a long beep to indicate that the pre-programmed time has been changed for the next incubation batch.

 **CAUTION:** After an incubation batch with a modified incubation time has finished, the timer setting returns to the pre-programmed time of 15 minutes.

6.6 Extending Incubator Useful Life

To extend the useful life of the DG THERM Incubator, turn it off at the end of the work day or when not in use.

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7 Maintenance

During the useful life under average conditions (as listed in Section 3.1), maintenance of the incubator is limited to regular cleaning and decontamination. After the useful life period, the incubator can still be used but, in addition to regular cleaning and decontamination, regular (at least yearly) and functional inspections are recommended to confirm that the thermal block temperature is within the required range, that the timers are accurate, and that the card elevator system is working properly. Perform these inspections by following the procedures listed in Section 8.

7.1 Maintenance Plan

The maintenance plan is a set of operations carried out at regular intervals to preserve the function of the DG THERM Incubator. As a general rule, and as long as there is no irregularity in the operation of the incubator, follow the maintenance plan listed in Table 3 to ensure the proper functioning of the incubator.

Table 3. Maintenance Schedule

PERIOD	ACTIVITY	CARRIED OUT BY
Whenever samples or other substances are spilled on the thermal block	Clean the thermal block and trays as described in Section 7.2.	OPERATOR
Monthly	Decontaminate the incubator as described in Section 7.3.	OPERATOR

Decontamination operations carried out by an operator or any other maintenance or repairs carried out by qualified personnel should be noted in Appendix A (included at the end of these Instructions for Use).

7.2 Cleaning the Incubator



WARNING: During cleaning and/or decontamination, wear protective gloves.



WARNING: Before disassembling, cleaning, or decontaminating the incubator, unplug the incubator from the power supply.



CAUTION: Before cleaning and/or decontaminating the incubator, remove all cards and/or sample tubes from inside the incubator.

Biological samples, saline solutions, acid, or alkaline solvents can cause damage to the exterior surface of the incubator and should be removed immediately. Periodically wipe the exterior surfaces with a cloth dampened with a soft detergent.

The trays should also be cleaned periodically. Pull them out and clean them with the same cloth used for cleaning the external surfaces. After cleaning, replace the trays.

- 
WARNING: If you spill cleaning solution or other liquids into the incubator, the incubator must be unplugged, cleaned, and decontaminated. The incubator must only be disassembled by qualified personnel and it must not be plugged in again until it has been fully inspected by qualified personnel.
- 
CAUTION: For the card elevator system to work properly, the trays must be completely and correctly loaded into the incubator.
- 
CAUTION: If the DG THERM Trays are damaged, replace them.

7.3 Decontaminating the Incubator

- 
WARNING: Any part of the incubator that comes into contact with blood, serum samples, or any other biological liquid should be treated as potentially infectious.
- 
WARNING: During cleaning and/or decontamination, wear protective gloves.
- 
WARNING: Before disassembling, cleaning, or decontaminating the incubator, unplug the incubator from the power supply.
- 
CAUTION: Before cleaning and/or decontaminating the incubator, remove all cards and/or sample tubes from inside the incubator.

To prevent risk of infection, decontaminate the incubator monthly, as well as in the following situations:

- After spills or leaks of potentially contaminating liquids.
- Before and after transport or storage.
- Before repairs are performed.
- Before incubator disposal.

To decontaminate the incubator, follow these instructions:

1. Turn off the incubator, disconnect the power cord, and wait until the incubator returns to room temperature.
2. In case of spills, wipe up the liquid with a disposable absorbent material (for example, paper towels, gauze, or cloth).

3. Wipe the entire incubator with disposable towels or cotton balls dipped in a 0.5% (v/v) aqueous solution of sodium hypochlorite until all the surfaces of the incubator are completely damp. Any other solution that is used must be validated by the individual site. The disposable towels or cotton balls should be dipped in the disinfectant solution until they are completely damp.
4. Rinse the surface with a soft detergent and water to eliminate odors and any residual solution.
5. Dry the surface.
6. Dispose of contaminated materials used during decontamination in a container for biological waste.



NOTE: The procedures described above do not guarantee that the incubator is completely decontaminated, but these procedures reduce the risk of contamination to a minimum.

7.3.1 Handling Decontamination Solution



WARNING: Some cleaning or decontamination products, such as sodium hypochlorite solution, may be corrosive, irritating for the skin and eyes, or toxic if inhaled, absorbed, or ingested.

During handling of the decontamination solution, follow these recommendations:

- These Instructions for Use.
- The laboratory's safety regulations.
- Any applicable local, state, or national regulations.

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8 Confirming Specifications (Operation Qualification Procedure)

The incubation temperature for the DG THERM Incubator has been calibrated by the manufacturer and incubation timing is controlled by two independent timers. If no function alarm is activated (see Sections 12.1 and 12.2), it is warranted that the incubator will incubate correctly for its useful life (see Table 1 in Section 3.1) without any additional calibration.

However, laboratories may need to periodically confirm that the DG THERM Incubator meets its specifications due to a quality management system or because they wish to use the incubator beyond its recommended useful life. To carry out these procedures, follow the instructions in Sections 8.2, 8.3, and 8.4 to confirm the following parameters:

- Incubation temperature.
- Incubation time.
- Card elevator system operation.

8.1 Required Materials

To confirm incubation temperature and timing, as well as the card elevator system, the following materials are required:

- PT100 temperature probe for DG Gel cards (Part # 230881).
- Digital thermometer for PT100 probes (Part # 230879).
- Calibrated chronometer (precision: ± 1 second).
- 12 DG Gel cards.

Calibrate the temperature probe and the thermometer using instruments controlled regularly with traceable materials at primary or secondary reference standards. To maintain traceability, the probe and the thermometer must have a label that includes the serial number, Calibration Certificate number, the calibration date, and the validity period.



NOTE: Any thermometer that meets the following specifications can be used:

- Compatible for PT100 Temperature probes (PT385 100 Ohm).
- Scale division (Resolution): ≤ 0.1 °C.
- Minimum range (°C): 20 to 40 °C.
- Maximum uncertainty permitted (thermometer plus probe): ± 0.3 °C (at 37 °C).
- Connection: IEC.

8.2 Confirming Incubation Temperature

To confirm that the thermal block is heating to the correct temperature as listed in Section 3.1, follow these instructions:

1. Unplug the incubator. Make sure that the incubator is at room temperature before proceeding with step 2.
2. Connect the temperature probe to the thermometer.
3. Insert the temperature probe in card position 6 of incubation area A of the thermal block. Make sure that at least 30 cm of the cable of the probe is in contact with the thermal block. Close the cover.
4. Plug in the incubator and turn it on. Make sure that the display is on and wait for the preheating phase to finish (after the preheating phase is complete, the timer will display the pre-programmed incubation time and the **Temperature Out-of-Range** symbol  disappears).
5. Turn on the thermometer.
6. Wait for 15 minutes and confirm that the thermometer temperature is within the temperature range defined in Section 3.1.
7. Confirm that the temperature that appears on the display of the incubator is also within the temperature range defined in Section 3.1.
8. Turn off the thermometer.
9. Open the cover and take out the temperature probe.



NOTE: If the incubation temperature measurement is not within the specifications listed in Section 3.1, repeat the test. If the results of the new test are not within the specifications, contact Technical Service.

8.3 Confirming the Timer

To confirm that the timers function correctly:

1. Ensure that the timer (1 or 2) shows the pre-programmed incubation time of 15 minutes.
2. Press the **Start/Stop Timer** button (the **P1** button or the **P2** button, depending on which timer is being tested) and start the chronometer at the same time.
3. Stop the chronometer at the same time as the incubation timer finishes. Confirm that the difference in the time between the incubator and the chronometer is less than 10 seconds.



NOTE: If the timer is not within 10 seconds of the chronometer timing, repeat the test. If the results of the new test are not within the specifications, contact Technical Service.

4. Repeat steps 1-3 for the second timer.

8.4 Confirming Card Elevator System Operation

To confirm that the card elevator system functions correctly:

1. Make sure that the incubator trays are completely loaded into their slots and correctly positioned.
2. Open the cover and load 12 DG Gel cards into incubation area A.
3. Confirm that all of the cards are elevated approximately 0.5 in (10 mm) above the thermal block.
4. Close the cover of incubation area A and confirm that the DG Gel cards are at the same height as the thermal block.
5. Open the cover of incubation area A again and confirm that all of the cards rise approximately 10 mm above the surface of the thermal block.
6. Remove the cards and close the cover of incubation area A.



NOTE: If the cards do not rise sufficiently when the cover of the incubator is opened or they are not at the same level as the surface of the thermal block when the cover is closed, make sure that the trays are correctly positioned. If the problem continues, contact Technical Service.

7. Repeat steps 1-6 for incubation area B.



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9 Transport and Storage



WARNING: Decontaminate the incubator before transport and/or storage.



CAUTION: Use only the original packaging to store or transport the incubator.

If you need to store the incubator for a long period of time, pack the incubator in the same manner as it was packed for transportation. Store the incubator in environmental conditions that meet the requirements listed in Section 3.1.

To store the incubator, you need a space with the following specifications:

- Minimum dimensions: 24 cm high by 40 cm deep by 40 cm wide (with packaging).
- Load-bearing capacity: 8 kg.



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10 Disposing of the Incubator

To dispose of the DG THERM Incubator, remove all remains of samples and reagents, and then clean and decontaminate it. Upon completion, dispose of the incubator as electronic waste, according to local, state, and national regulations.

If you have any questions about the disposal of the incubator, contact Technical Service.



WARNING: The incubator contains electronics. Electronics can contain hazardous materials. Dispose of all electronics in accordance with local, state, and national regulations.

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11 Warranty

The warranty conditions are established in separate documents. For more information, contact Technical Service.

The warranty does not cover:

- Damage caused by using the incubator in environmental conditions outside of the conditions listed in Section 3.1.
- Damage caused by accident, negligence, or breach of these Instructions for Use.
- Damage caused for reasons beyond our control (atmospheric, geological phenomena, etc.).
- Damage caused by incorrect handling, including handling by unauthorized personnel.
- The plastic or rubber components, as well as the enamel or paint, that have deteriorated due to normal usage, except in the case of a manufacturer defect.
- Damage to the trays, except in the case of a manufacturer defect.
- Damage that occurs during transport.

The replacement of equipment or components will be subject to the criteria of Diagnostic Grifols S.A.



CAUTION: Repairs should only be carried out by qualified personnel.

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12 Errors and Alarms

An operator can resolve most of the error messages and alarms that may occur during normal operation of the DG THERM Incubator. The following sections describe possible error situations and the steps to resolve them, as well as the situations that might activate the alarm.

12.1 Clearing Errors

Table 4 lists the error symbols that the incubator displays in certain error situations, along with a description of the error and the steps to take to try to resolve the error.

Table 4. Error Symbols

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
--:--	Incubator in preheating phase	The symbols “--:--” indicate that the incubator is in the preheating phase and not yet ready to carry out an incubation batch because the thermal block has not reached the work temperature yet. Once incubation temperature has been reached, the  disappears and the timers show the pre-programmed incubation time of 15 minutes.

ERROR SYMBOL	ERROR MESSAGE	DESCRIPTION
	Temperature out-of-range	The temperature sensors detect when the temperature of the thermal block is outside the established range and prevent the incubator timers from starting. Confirm that: - The incubator is not in preheating phase (display showing “--:--”). If the temperature of the thermal block is outside the established margins after this phase, the alarm beeps until a key is pressed. - The power to the incubator has not been cut off. If the power has been cut off, the timers flash the time remaining and the alarm beeps. After the power comes back on, the incubator goes into preheating phase but the alarm continues to beep and the timers continue to flash. If the Temperature Out-of-Range symbol  is still displayed after the preheating phase, turn the incubator off and back on again. If the error symbol is still displayed after the preheating phase, contact Technical Service.
	Erroneous incubation time	The timing of a batch was different from the pre-programmed time and the corresponding timer is flashing. This may be due to: - Timing being stopped by the operator as described in Section 6.4. The timer flashes the time remaining. - A loss of power. The timer flashes the time remaining and the alarm beeps. Press the corresponding Start/Stop Timer button (the P1 button or the P2 button) to stop the alarm and reset the timer. Discard the DG Gel cards and invalidate the test results.
	Contact Technical Service	If the incubator detects any alteration in its function that cannot be resolved by the operator, the Repair symbol  appears. Turn the incubator off and back on again. If the error symbol appears again, contact Technical Service.

If the incubator detects any error conditions other than those errors already mentioned, the incubator will:

- Stop heating the thermal block.
- Not respond to the **P1** button or the **P2** button.
- Activate the alarm.
- Display the **Repair** symbol , along with an **E** and a corresponding error code.



CAUTION: If the DG THERM Incubator displays an error code, turn the incubator off and then back on again. If the error code appears again, contact Technical Service and give them the error code.

12.2 Alarm Situations

The DG THERM Incubator has an alarm with three volume level settings. The alarm sounds in the following situations:

- The correct incubation temperature is reached. A short beep indicates that the preheating phase has finished, the **Temperature Out-of-Range** symbol  disappears, and the timers display the pre-programmed incubation time of 15 minutes.
- The incubation batch is finished. The alarm emits a long beep.
- Incubation timing has been stopped by pressing and holding the corresponding **Start/Stop Timer** button. The alarm emits three short beeps and a continuous beep.
- A loss of power has occurred during an incubation batch. When the power returns, the alarm emits a continuous beep and the timer flashes the time remaining from the time of the power outage.
- The temperature of the thermal block is outside the established range after the preheating phase. The alarm emits a long beep and the timer flashes the time remaining.
- An error has occurred during incubation. The alarm beeps and the corresponding error symbol is displayed.



CAUTION: To stop the alarm, press the **P1** button or the **P2** button. If the alarm continues, contact Technical Service.



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Maintenance Log

DATE	ACTION CARRIED OUT	PERFORMED BY & DATE	SUPERVISOR & REVIEW DATE